



United International University

B.Sc. in Computer Science & Engineering (CSE)

CSE 4889: Machine Learning

Midterm Exam (Set A): Spring 2025 Time: 1 Hour 30 Minutes Marks: 30

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Answer all of the following questions.

1. a) The class data are given in Table 1. Classify the new data point (5.10, 3) using [6]
Mahalanobis distance.

Table 1

Data	Class
(2.5, 3.25)	Class-I
(4.5, 1.5)	Class-I
(2.0, 3.0)	Class-II
(4.0, 1.0)	Class-II
(3.20, 2.5)	Class-I
(3.0, 2.0)	Class-II

2. a) The dataset is given as (4, 3), (6, 7), (3, 5) and (4, 5). The cluster number and [6]
fuzziness coefficient are $C=2$ and $m=3$, respectively. Evaluate $|U_{32}^{(1)} - U_{32}^{(0)}|$ by
applying appropriate clustering method where termination criterion is 0.2.
- b) Specify the significant difference between K-means and Fuzzy C-means [2]
clustering methods concerning clustered outcomes.
3. a) A dataset contains information on customer spending in a retail store with the [4]
following features.

Table : 2

ID	Age	Income (k\$)	Spending score	Label
1.	25	30	45	Low
2.	35	60	80	High
3.	23	45	50	Medium
4.	45	70	85	High
5.	30	40	55	Medium
6.	55	85	90	High

Apply the k-nearest neighbor algorithm with $k=5$ to classify the new customer for
the features: Age=50, income=65000 \$, Spending score= 70.

b) Let us consider the data points as shown in Table 3. Identify **core**, **outlier** and **border** through proper density based clustering technique. Similarity index $\varepsilon = 1.89$ (maximum) and minimum points = 3. [4]

Table : 3

P1	(3, 7)
P2	(4, 6)
P3	(5, 5)
P4	(6, 4)
P5	(7, 3)
P6	(6, 2)

4. a) Apply an appropriate supervised classification approach to decide whether a student will play or not using the tuple **X** (temperature=medium, student =Yes, Air quality= Fair) for the following dataset. [8]

Table : 4

ID	Temperature	Hungry	Student	Air quality	Play
1.	Low	No	No	Fair	No
2.	Low	Yes	No	Excellent	No
3.	Medium	Yes	No	Fair	Yes
4.	High	No	No	Fair	Yes
5.	High	Yes	Yes	Fair	Yes
6.	High	Yes	Yes	Excellent	No
7.	Medium	Yes	Yes	Excellent	Yes
8.	Low	No	No	Fair	No
9.	Low	Yes	Yes	Fair	Yes
10.	High	Yes	Yes	Fair	Yes
11.	Low	Yes	Yes	Excellent	Yes
12.	Medium	No	No	Excellent	Yes
13.	Medium	Yes	Yes	Fair	Yes
14.	High	No	No	Excellent	No